1. Objective

The objective of this project is to identify users most likely to respond to an invitation to join the rewards program, using targeted perks to maximize adoption. The focus is not on increasing overall activity, but on understanding behavioral patterns to align incentives with user needs.

2. Data Preparation

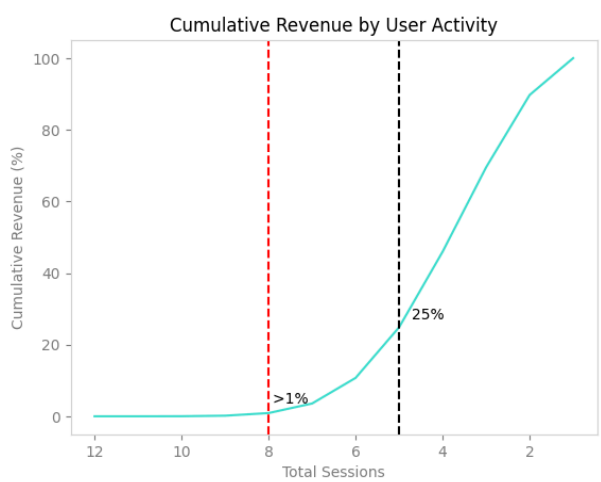
The analysis began with the construction of a trip-level dataset by combining session, flight, hotel, and user data. Several inconsistencies were identified, including negative trip durations and misaligned timestamps between flights and hotel stays.

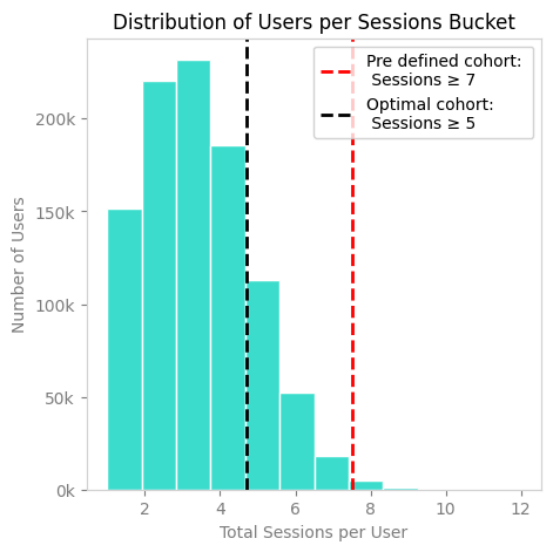
Rather than correcting these values artificially, invalid records were removed to preserve data integrity. These cases represented a small proportion of the dataset and were treated as data quality issues rather than valid edge cases.

A user-level dataset was then created by aggregating behavioral, transactional, and demographic features such as total sessions, conversion rate, revenue, cancellation rate, and travel characteristics.

3. Cohort Definition

To focus the analysis on relevant users, a cutoff of sessions ≥ 5 was applied. This threshold captures approximately 25% of the user base while accounting for around 25% of cumulative revenue. This ensures that the analysis targets users with meaningful engagement without excluding a significant portion of business impact.





4. Rule-Based Segmentation

Three initial segments were defined based on clear behavioral patterns:

- No Purchase Users: users with sessions but no bookings or revenue
- Trips Without Revenue: users who completed bookings but generated no revenue, typically due to cancellations
- High Value Users: top 5% of users by total revenue

These segments represent distinct behavioral extremes and were separated prior to clustering to avoid distortion in the model.

5. Clustering Approach

The remaining users were segmented using unsupervised machine learning. Multiple approaches were tested, including clustering on raw features, engineered indices, and PCA-reduced datasets.

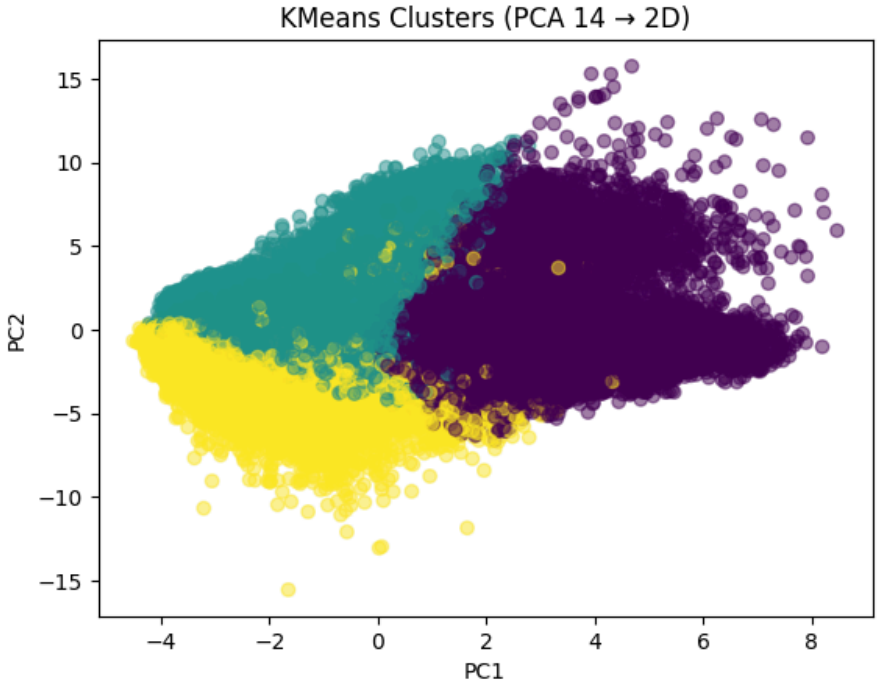
While higher-dimensional PCA preserved more variance, it resulted in less interpretable clusters. A 2-component PCA provided clearer separation of user behavior and enabled more meaningful segmentation. KMeans was selected as the final algorithm due to its stability and interpretability.

6. Clustering Results

Three behavioral clusters were identified:

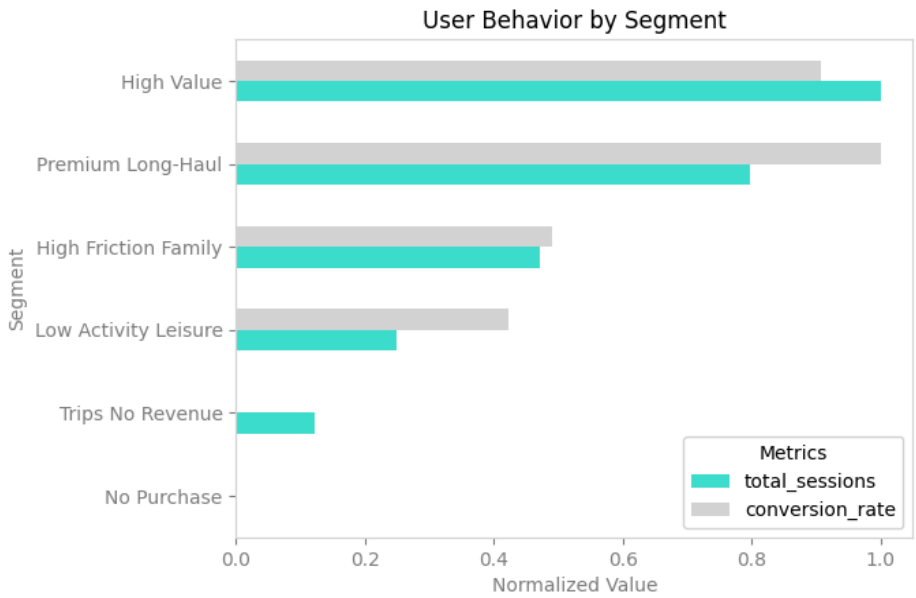
- **Premium Long-Haul Travelers:** high revenue, high activity, long-distance trips, and consistent behavior
- **High-Friction Family Travelers:** moderate activity, high cancellation rates, and family-oriented travel patterns
- **Low-Activity Leisure Travelers:** lower engagement, shorter distances, and longer, more stable trips

The results show that activity alone does not differentiate users. Instead, conversion behavior, cancellation patterns, and travel intent are the key drivers.



7. Key Insights

- High activity does not guarantee high conversion or revenue
- Cancellation behavior is a strong indicator of friction and uncertainty
- Travel purpose (e.g., family vs individual) significantly impacts behavior
- A small subset of users contributes disproportionately to value



8. Business Recommendations

The segmentation aligns closely with the proposed rewards program incentives:

- Free hotel meals are well suited for leisure travelers seeking experience-based value
- Free checked bags address the logistical needs of family travelers
- No cancellation fees directly reduce friction for users with repeated cancellations
- Exclusive discounts are effective for users who have not yet converted
- A free hotel night bundled with flights enhances perceived value for premium travelers
- Cashback or rewards-based incentives reinforce loyalty among high-value users

These targeted incentives are expected to increase rewards program adoption by addressing specific user needs rather than applying a generic approach.

9. Conclusion

This project demonstrates that effective segmentation requires both data-driven methods and business context. By combining rule-based segmentation with machine learning, the analysis identifies clear user groups and aligns them with actionable strategies.

The resulting framework provides a scalable foundation for personalized marketing and improved user engagement within the rewards program.